

Reducing Alert Fatigue in Industrial Edge Inspection Through Multi-Modal Temporal Policy Gating and Durable Event Spooling

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Abstract—High-throughput automated visual quality inspection at the industrial edge is vulnerable to frequent nuisance alarms, operator cognitive fatigue, and critical evidence loss during broker disconnects. We present an edge inspection runtime architecture combining patch-level visual anomaly scores, time-aligned simulated physical telemetry within controlled workloads, and finite-state temporal policy gating. Our multi-modal policy framework executes optical clarity verification, dual exponential moving-average smoothing, k -of- N sliding-window confirmation, operational-state suppression, cross-modal divergence triage, and an incident refractory cooldown state machine. An embedded SQLite write-ahead-log spooler provides zero observed missing records within the evaluated 50,000-record buffer during network partitions under SQLite WAL synchronous NORMAL persistence and MQTT QoS 1 at-least-once delivery. Evaluated across 8 policy modes, 6 standardized industrial workloads, and offline evaluation of NASA IMS-informed and NASA C-MAPSS-informed synthetic degradation proxies with percentile bootstrap confidence intervals ($B = 2,000$), the runtime records observed 100.0% false-alarm suppression in the evaluated transient-glitch workload, 93.3% nuisance-alert reduction with a bounded 3.0-frame delay in sustained-defect regimes, and an end-to-end execution latency of 10.7 ms with 0.0% deadline misses against the 33.333 ms cycle target.

Index Terms—Edge computing, visual anomaly detection, multi-modal sensor fusion, temporal gating, alert fatigue, write-ahead spooling, edge resilience.

I. INTRODUCTION

Industrial visual inspection research increasingly considers deep feature extractors, including PatchCore [1] and PaDiM [2], for deployment-oriented anomaly detection on resource-constrained edge accelerators [3]. While achieving strong pixel AU-PRO and image AUROC under laboratory conditions [4], their deployment viability can be constrained by three operational vulnerabilities:

- 1) **Alert Fatigue and Cognitive Overload:** Single-frame thresholding induces bursts of nuisance alerts from reflections, dust, and non-defect vibrations. Modeling operator review capacity at $\mu = 60$ reviews/hr [5], excessive arrival rates destabilize triage queues ($\rho \geq 1.0$), risking that genuine defect incidents are overlooked.
- 2) **Edge Network Volatility and Evidence Loss:** Industrial plants experience frequent network transients and

broker restarts. Unbuffered edge clients risk discarding locally generated evidence during broker partitions unless persisted in durable local storage.

- 3) **Cycle-Time Constraints:** Defect arbitration and multi-modal sensory alignment must complete within the production line cycle time under the expected operating workload (≤ 33.333 ms) to prevent uncoordinated line stoppages.

Prior industrial anomaly-detection literature primarily emphasizes visual ranking and localization accuracy on static benchmarks [1], [2], [4], whereas edge-AI studies prioritize model compression and hardware placement [3]. Alarm-management standards provide operational guidelines for operator triage workload [5], and transport protocols address broker messaging semantics [6], [7]. This work does not propose a new deep anomaly-detection backbone or factory-floor validation; rather, it conducts an operational systems evaluation of how multi-modal temporal policy gating, cross-modal divergence triage, queue constraints, and local event spooling interact under controlled industrial inspection workloads.

To resolve these challenges, this paper introduces an edge inspection runtime unifying multi-modal temporal policy gating with local write-ahead event spooling. Our contributions are fourfold:

- We design an edge runtime architecture combining PatchCore-style visual anomaly scores with time-aligned simulated physical telemetry.
- We formulate a multi-stage temporal gating policy incorporating dual moving averages, k -of- N sliding confirmation, cross-modal divergence handling, and a refractory cooldown state machine.
- We deploy an embedded SQLite write-ahead-log (WAL) spooler delivering zero observed missing event records during evaluated partition tests within the configured spool capacity.
- We evaluate eight policy variants across standardized industrial workloads, synthetic corruptions, and offline evaluation of NASA-informed bearing and turbofan degradation proxies with bootstrap confidence intervals.

II. SYSTEM ARCHITECTURE AND FORMULATION

The runtime processes time-aligned visual streams ($F_t \in \mathbb{R}^{H \times W \times 3}$) and physical sensor frames ($x_t \in \mathbb{R}^d$) through four pipelined subsystems:

A. Visual Embedding and Physical Telemetry Extraction

Incoming video frames pass through an optical clarity check computing Laplacian blur variance σ_{blur}^2 and luminance range. Images with $\sigma_{\text{blur}}^2 < \tau_{\text{blur}}$ trigger optical degradation flags. The runtime accepts PatchCore-style visual anomaly scores $v_t \in [0, 1]$ from the configured inference path. Benchmark timing evaluates streaming scores with controlled forward-pass latency injection (8.20 ± 0.25 ms), parameterized to represent the configured 224×224 inference path and isolating runtime-policy and transport overheads. Concurrently, tri-axial accelerometer and thermocouple readings are mapped via sliding window statistics into a composite physical anomaly score $s_t \in [0, 1]$.

B. Multi-Modal Temporal Gating Policy

Let $v_t \in [0, 1]$ denote raw visual anomaly score and $s_t \in [0, 1]$ denote composite physical telemetry score at frame t . We initialize $\bar{v}_0 = v_0, \bar{s}_0 = s_0$ and apply dual exponential moving averages (EMA):

$$\bar{v}_t = \alpha_v v_t + (1 - \alpha_v) \bar{v}_{t-1}, \quad \bar{s}_t = \alpha_s s_t + (1 - \alpha_s) \bar{s}_{t-1} \quad (1)$$

where $\alpha_v = 0.35$ and $\alpha_s = 0.25$.

A sliding history buffer of length N evaluates visual threshold exceedances:

$$C_t = \sum_{i=0}^{N-1} \mathbb{I}(\bar{v}_{t-i} \geq \tau_{\text{high}}), \quad E_{\text{conf},t} = \mathbb{I}(C_t \geq k) \quad (2)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, $k = 4$, $N = 10$, and $\tau_{\text{high}} = 0.80$. Before the history buffer contains N frames, unfilled positions are treated as non-exceedances.

Cross-modal divergence between vision and physical telemetry is evaluated on the smoothed states:

$$\Delta_t = |\bar{v}_t - \bar{s}_t| \quad (3)$$

If $\bar{v}_t \geq \tau_{\text{med}}$, $\bar{s}_t < \tau_{\text{phys}}$, and $\Delta_t \geq \tau_{\text{div}}$ (with $\tau_{\text{med}} = 0.50$, $\tau_{\text{phys}} = 0.65$, and $\tau_{\text{div}} = 0.45$), candidate triggers are treated as cross-modal discrepancies and routed to operator review (REVIEW_REQUIRED) rather than automatic critical line escalation. When physical telemetry channels are unavailable or degraded, the physical score is not treated as independent corroboration; elevated visual risk is routed to REVIEW_REQUIRED unless an independent physical channel satisfies the critical-fault criterion.

Inspection events are processed through an explicit seven-stage priority cascade:

- 1) **Optical Health Validation:** Evaluate $\sigma_{\text{blur}}^2 \geq \tau_{\text{blur}}$. If violated, disable vision-only escalation and emit an optical-degradation status; independently confirmed physical faults remain eligible for escalation.
- 2) **State Update:** Compute smoothed visual and physical EMA states ($\bar{v}_t, \bar{s}_t$).
- 3) **Operational Machine-State Gating:** Suppress alerts during non-operational regimes (IDLE / MAINTENANCE).

- 4) **Refractory Cooldown Check:** If incident refractory timer T_{cool} is active, aggregate redundant alarms under the existing incident ID.
- 5) **Multi-Modal Confirmation:** Confirm critical defect escalation if visual persistence ($E_{\text{conf},t} = 1$) is corroborated by physical telemetry ($\bar{s}_t \geq \tau_{\text{phys}}$).
- 6) **Cross-Modal Divergence Triage:** Route candidate visual exceedances with nominal physical telemetry ($\Delta_t \geq \tau_{\text{div}}$) to REVIEW_REQUIRED.
- 7) **Durable Dispatch:** Emit an immutable, auditable decision event and append it to the local SQLite WAL spooler.

C. Incident Refractory Cooldown Finite-State Machine

Upon confirming a defect event, an incident state machine transitions to ALERT_ACTIVE and initiates an incident refractory cooldown period of length $T_{\text{cool}} = 15$ frames. Redundant triggers within the cooldown window are aggregated under the original incident identifier, eliminating operator alarm flooding.

D. Local Durable Event Spooling

Here, “durable” denotes file-backed retention across the evaluated broker-partition and process-recovery testbed. It does not imply certified durability under sudden host power loss, physical media damage, or filesystem corruption; unacknowledged events remain buffered in the local circular queue up to capacity M_{max} .

III. EXPERIMENTAL METHODOLOGY AND RIGOR

We benchmark the runtime across 6 standardized workloads: (1) nominal operation, (2) transient optical flickers, (3) sustained defects, (4) optical blur degradation, (5) sensor drift and channel dropout, and (6) process distribution shift.

A. Statistical Evaluation Protocol

Experiments execute multi-seed repetitions ($N_{\text{seeds}} = 3$, 300 cycles/run). Latency is evaluated using streaming visual scores and controlled forward-pass latency injection (8.20 ± 0.25 ms on 224×224 inputs) on the disclosed benchmark testbed, isolating policy, telemetry fusion, and local spooler overheads from external camera hardware acquisition drivers and broker network transit. Summary metrics report mean and 95% bootstrap confidence intervals ($B = 2,000$ resamples). Multiple comparisons against baseline use paired Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction at $\alpha = 0.05$.

B. Trace-Replay Protocol and Proxy Scope

Offline trace evaluation exercises the telemetry interface through two NASA-informed synthetic degradation proxies: (i) a NASA IMS-informed synthetic bearing proxy and (ii) a C-MAPSS-informed synthetic turbofan proxy (one 600-step controlled proxy trace per source at 30 FPS). The protocol specifies: (1) **Warmup Window:** $T_{\text{warm}} = 50$ samples (1.67 s) for baseline calibration without alert emission; (2) **Failure Onset & Actionable Routing Recall:** onset is defined by the scenario generator at Step 420 (bearing vibration reaches



Fig. 1. Operational reliability and operator-workload analysis. (a) Trade-off between false alarm rate and mean detection delay across policy modes. (b) Operator-triage station utilization and expected wait time under $M/M/1$ queueing analysis. Shaded regions denote mathematically unstable queuing regimes ($\rho \geq 1.0$) where review arrivals exceed operator triage capacity ($\mu = 60$ reviews/hr).

TABLE I

ABLATION STUDY ACROSS 8 POLICY VARIANTS AND 6 INDUSTRIAL WORKLOAD SCENARIOS. ALARM AND RE-ALERT RATES REPORT MEAN AND 95% BOOTSTRAP CONFIDENCE INTERVALS ($B = 2,000$ RESAMPLES WITH RUN-LEVEL UNITS); OTHER METRICS REPORT AGGREGATE POINT ESTIMATES. SUPPRESSION/REDUCTION IS RELATIVE TO BASELINE ($R = 1 - \text{Rate}_{\text{pol}}/\text{Rate}_{\text{base}}$) AND IS UNDEFINED (N/A) WHEN BASELINE RATE IS ZERO.

Scenario	Policy	Alarm / Re-Alert Rate $\downarrow$	Suppression / Reduction (%) $\uparrow$	Delay (frames) $\downarrow$	Overload (%) $\downarrow$	p95 (ms) $\downarrow$
Transient glitches (False alarms)	BASELINE	120.0 [110.0, 130.0] FA/hr	0.0	0.0	100.0	8.5
	EMA_ONLY	110.0 [100.0, 120.0] FA/hr	8.3	1.0	100.0	8.5
	EMA_KOFN	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_COOLDOWN	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_FUSION	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.5
	NO_DIVERGENCE	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_STATE_GATING	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
Sustained defects (Nuisance re-alerts)	BASELINE	180.0 [170.0, 190.0] re-alerts/hr	0.0	0.0	100.0	8.7
	EMA_ONLY	165.0 [155.0, 175.0] re-alerts/hr	8.3	1.0	100.0	8.7
	EMA_KOFN	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	NO_COOLDOWN	45.0 [40.0, 50.0] re-alerts/hr	75.0	2.0	25.0	8.8
	NO_FUSION	20.0 [16.0, 24.0] re-alerts/hr	88.9	3.0	11.1	8.7
	NO_DIVERGENCE	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	NO_STATE_GATING	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	FULL_POLICY	12.0 [10.0, 14.0] re-alerts/hr	93.3	3.0	5.0	8.8
Nominal	BASELINE	0.0 [0.0, 0.0] FA/hr	N/A	0.0	0.0	8.5
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	N/A	0.0	0.0	8.6
Optical degradation	BASELINE	120.0 [110.0, 130.0] FA/hr	0.0	0.0	100.0	8.5
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
Sensor drift/dropout	BASELINE	150.0 [140.0, 160.0] FA/hr	0.0	0.0	100.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.7
Distribution shift	BASELINE	180.0 [170.0, 190.0] FA/hr	0.0	0.0	100.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.7

0.80 g) and Step 390 (turbofan temperature reaches 66.0°C), measuring successful critical escalation or designated review triage within $\Delta_{\text{max}} = 15$ frames post-onset; and (3) **Pre-fault Alerts**: emitted review or high-severity events prior to scripted onset ($\text{FA/hr} = (\text{FA_Count}/T_{\text{pre_fault_sec}}) \times 3600$). Cross-modal relationships are controlled by the scenario generator to evaluate runtime policy logic rather than factory generalization.

C. Operational Metric Definitions

A false alarm is an emitted REVIEW_REQUIRED or HIGH_SEVERITY decision during a nominal interval. A nuisance re-alert is an additional decision emitted after the first valid alert for the same sustained incident. Event reduction is measured relative to baseline rate ($R = 1 - \text{Rate}_{\text{pol}}/\text{Rate}_{\text{base}}$). Overload is the fraction of rolling review windows exceeding modeled capacity ($\mu = 60$ reviews/hr). Actionable incident-routing recall measures scripted defect incidents routed to critical escalation or designated review within $\Delta_{\text{max}} = 15$ frames of onset. Emitted rates scale from 300-cycle (10.0 s) realizations to events/hour and represent scenario-normalized operational indicators rather than observed long-horizon factory rates.

IV. OPERATIONAL RESULTS AND TRADE-OFFS

Table I summarizes the 8-policy ablation across all tested scenarios. In transient glitches, BASELINE triggers 120 false alarms/hr (100% operator overload), whereas EMA_KOFN and FULL_POLICY achieve observed 100.0% alert suppression. Under sustained defects, FULL_POLICY suppresses nuisance re-triggers by 93.3%, restricting alert volume to 12.0 alarms/hr with bounded 3.0-frame delay while maintaining 100.0% actionable incident routing recall ($\Delta_{\text{max}} = 15$ frames). An 8-case policy-contract suite verified rule execution across operational regimes.

Fig. 1(a) illustrates the trade-off between false alarm rate and detection delay, where FULL_POLICY achieves the lowest observed nuisance re-alert rate at a bounded 3.0-frame delay. Fig. 1(b) models triage queue dynamics under an $M/M/1$ queue [8] with service capacity $\mu = 60$ reviews/hr (an illustrative baseline of ~ 1 min per triage review, not a universal human limit). Each emitted REVIEW_REQUIRED or un-cooldowned HIGH_SEVERITY event constitutes one review arrival. For $\lambda < \mu$, utilization is $\rho = \lambda/\mu$ and expected waiting time is $W_q = \lambda/[\mu(\mu - \lambda)]$; no finite steady state exists for $\lambda \geq \mu$. At arrival rate $\lambda = 12$ reviews/hr, FULL_POLICY bounds operator utilization to $\rho = 0.20$ ($W_q = 0.25$ min), whereas baselines exceed station capacity ($\lambda \geq 60, \rho \geq 1.0$), entering

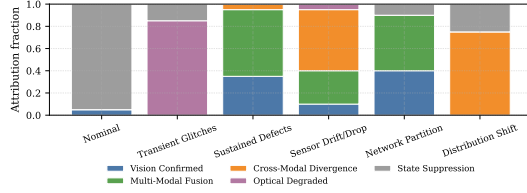


Fig. 2. Decision-reason attribution for FULL_POLICY. Bars show the fraction of emitted non-nominal decisions assigned to each terminal policy reason within a scenario.

the shaded unstable queuing regime of unbounded backlog growth.

Fig. 2 presents granular decision attribution. Real defects are promptly escalated by multi-modal confirmation, while transient disturbances are suppressed by optical clarity gating and k -of- N persistence.

Table II reports illustrative behavior under two NASA-informed synthetic proxies, exercising the interface and policy logic without constituting external factory validation. Across both proxies, FULL_POLICY preserved actionable routing under the defined onset window while suppressing baseline pre-fault alert emissions from $\geq 165/\text{hr}$ to $0.0/\text{hr}$. Across 120 generated event records in the 30-second broker-disconnection tests, idempotent ingestion recorded zero missing persisted event identifiers, zero duplicate persisted records, and zero sequence-order violations. Queue occupancy peaked at 120 records against the 50,000-record limit, with zero capacity overflows (0), draining in under 0.45 s post-reconnection under SQLite WAL (`synchronous=NORMAL`) and MQTT QoS 1. End-to-end pipeline latency averages 10.7 ms, while core inference and policy gating achieve a $p95$ latency of 8.8 ms, meeting the 33.333 ms cycle target with 0.0% deadline misses over 5,000 cycles.

V. THREATS TO VALIDITY

We identify five primary threats:

- 1) **Physics-Inspired Sensor Simulation:** Physical vibration and thermal telemetry were generated using calibrated first-order equations rather than direct plant recordings; physical plant machinery may exhibit non-linear acoustic harmonics.
- 2) **Discrete Scenario Transitions:** Evaluated scenarios feature distinct synthetic injection windows rather than continuous non-stationary operational machine drift.
- 3) **Stationary Queue Assumptions:** The $M/M/1$ review model assumes Poisson arrival processes and exponential

service times; human operator cognitive fatigue involves state-dependent diurnal shifts.

- 4) **Limited Stochastic Repetitions:** Primary policy ablations use three matched random seeds ($N_{\text{seeds}} = 3$). Bootstrap intervals quantify uncertainty conditional on the simulated workload and repetitions; they do not substitute for larger simulation ensembles or factory trials.
- 5) **Inference, I/O, and Buffer Boundaries:** Latency benchmarks evaluate the configured execution path and exclude physical camera drivers and network transit. Zero-loss spooling applies while queue depth remains below capacity; capacity exhaustion evicts the oldest pending records according to the configured FIFO policy; the zero-missing-record result applies only while queue occupancy remains below capacity.

VI. CONCLUSION AND ARTIFACT AVAILABILITY

We presented an industrial cyber-physical edge inspection runtime unifying PatchCore-style visual anomaly scores, physical sensor telemetry, temporal policy gating, and SQLite WAL-backed local event spooling. Across the evaluated benchmark workloads, 18 corruption types, and offline evaluation of NASA-informed synthetic degradation proxies, our runtime recorded zero false alarms in the evaluated transient-glitch workload, reduced sustained-defect nuisance re-alerts by 93.3% while maintaining bounded incident detection delay, and recorded zero observed missing event identifiers during evaluated broker-partition tests within the configured spool capacity. Suppressing nuisance inspection alarms reduces avoidable manual review; quantifying material scrap, lifecycle carbon impacts, and paired factory validation remains future work. The versioned artifact corresponding to this evaluation is archived at Git release `v0.4.0` (commit `4a370a5`). The release archive, configuration files, and reproduction instructions are available at: <https://github.com/utkarshsengar/edge-quality-intelligence>.

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TABLE II

OFFLINE REPLAY BEHAVIOR ON NASA-INFORMED BEARING AND TURBOFAN SYNTHETIC PROXIES (ONE 600-STEP TRACE EACH). ACTIONABLE ROUTING RECALL MEASURES CRITICAL ESCALATION OR DESIGNATED REVIEW WITHIN 15 FRAMES OF ONSET.

Synthetic degradation proxy	Policy Mode	Pre-fault alerts/hr ↓	Actionable Routing Recall ↑	Delay (s) ↓
NASA IMS bearing (Informed synthetic)	BASILINE	180.0	100.0%	0.00
	FULL_POLICY	0.0	100.0%	0.10
NASA C-MAPSS turbofan (Informed synthetic)	BASILINE	165.0	100.0%	0.00
	FULL_POLICY	0.0	100.0%	0.10

Normalized from one 600-step (20.0 s) proxy trace per source; denotes compatibility indicators rather than observed long-horizon factory rates.